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Large-Scale Optimization for Recommender Systems

Matrix Completion for Recommender Systems: Evaluating Standard, Away-Step, and Pairwise Frank-Wolfe Algorithms

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Note: Generative AI tools were utilized exclusively as an individual study aid, specifically to understand the theoretical and geometric implications of optimizing over non-polyhedral domains such as the nuclear norm ball, as well as to assist with grammatical review. All experimental designs, code implementations, and final analyses are the original work of the authors.

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Abstract

The following report analyzes the application of the Frank-Wolfe algorithm and its variants with the aim of solving the Matrix Completion problem in Recommender Systems. Due to the computational intractability of the rank constraint, the problem is addressed using a convex relaxation based on the nuclear norm. The computational advantage provided by utilizing a Linear Minimization Oracle compared to traditional projection methods is analyzed, leveraging truncated SVD computations, that extract only the leading singular vector pair. Finally, the report investigates the geometric limitations of the standard Frank-Wolfe algorithm, particularly the zigzagging phenomenon, and explores how its advanced variants, such as the Away-Step and Pairwise Frank-Wolfe algorithms, overcome these structural bottlenecks trying to accelerate the convergence.

1 Introduction

Modern recommender systems rely on the analysis of user-item matrices to extract latent behavioral patterns and preferences. These matrices are typically highly sparse, since only a small fraction of possible user-item interactions are observed. The analysis is based on the unsupervised learning model known as Matrix Completion, which aims to predict missing entries in order to suggest relevant new content. The rationale behind this assumption is that user's choices are driven by a limited number of latent factors; this translates into the search for a low-rank matrix capable of accurately approximating the known observations to successfully predict the missing entries. Although the original formulation requires the direct minimization of the matrix rank, this operation is classified as an NP-hard problem, making it necessary the usage of convex relaxation techniques. Classical constrained optimization algorithms, such as Projected Gradient Descent (PGD), require solving complex quadratic subproblems to project the solution back into the feasible space. In the context of matrices, this consist in computing a full Singular Value Decomposition (SVD) at each iteration; in order to overcome this computational impediment, the Frank-Wolfe (FW) method, introduced in 1956 by Marguerite Straus-Frank and Philip Wolfe, represents a highly scalable alternative[1]. The algorithm belongs to the class of projection-free first-order methods, replacing expensive exact projections with the resolution of a Linear Minimization Oracle (LMO) over the domain[1]. Because the LMO is often significantly cheaper to compute than a projection, the method naturally generates sparse, structured approximations of the optimal solution[1].

2 Mathematical Formulation

Consider a partial sparse feedback matrix $Y \in \mathbb{R}^{n_1 \times n_2}$, where n_1 represents the number of users and n_2 the number of items. Let Ω define the set of indices for which an observed rating exists. The ideal matrix completion problem requires finding a matrix X of minimum rank that strictly respects the following observation:

$$\min_{X \in \mathbb{R}^{n_1 \times n_2}} \operatorname{rank}(X) \quad \text{s.t.} \quad \mathcal{P}_\Omega(X) = \mathcal{P}_\Omega(Y) \tag{1}$$

where $\mathcal{P}_\Omega(\cdot)$ is the orthogonal projection operator onto the set of observed indices. Since the rank-function is non-convexity, the problem is reformulated by minimizing a smooth, differentiable objective function $f(X)$ subject to a convex constraint. The loss function employed is the half sum squared error over the known entries:

$$f(X) = \frac{1}{2}\|\mathcal{P}_\Omega(X - Y)\|_F^2 \tag{2}$$

To bypass the intractable rank constraint while still inducing a low-rank structure, the original constraint is replaced by a restriction on the nuclear norm of the matrix, which is defined as the sum of its singular values ($\|X\|_* = \sum_i \sigma_i(X)$).
The resulting convex optimization problem takes the form:

$$\min_{X \in \mathbb{R}^{n_1 \times n_2}} f(X) \quad \text{s.t.} \quad \|X\|_* \leq \tau \tag{3}$$

where $\tau > 0$ is a regularization hyperparameter defining the radius of the nuclear norm ball. It serves as a balance between fitting the training data and tolerating noise. Consequently, the optimization domain is the convex and compact set $\mathcal{C} = \{X \in \mathbb{R}^{n_1 \times n_2} : \|X\|_* \leq \tau\}$.

3 Experimental Setup

3.1 Datasets and Data Preprocessing

To evaluate the performance of the Frank-Wolfe algorithm and its variants on real-world tasks, the experiments have been conducted using three different datasets:

- **MovieLens 100k:** A standard collaborative filtering benchmark containing user ratings for various movies[2].
- **Amazon Video Games:** A sparse product review dataset extracted from Amazon[3].
- **MyAnimeList:** A user-rating dataset reflecting preferences across anime titles[4].

To ensure a robust evaluation, data preprocessing has been applied by filtering out users with fewer than a minimum threshold of ratings[5]. User and item identifiers are subsequently mapped to contiguous 0-indexed integer coordinates to construct the sparse matrix structures[5]. Each dataset was partitioned into an 80% training set, a 10% validation set, and a 10% hold-out test set, stratified by user index to preserve rating distributions[5].

Dataset Structure Analysis. A detailed analysis of the datasets is necessary to better understand the final results achieved. Table 1 summarizes the sparsity characteristics of the three training sets after preprocessing. The three datasets show different density profiles: Amazon Video Games is the sparsest (99.52%), with an average of only 4.6 ratings per item, while MyAnimeList is the densest (87.63%), with an average of 100.6 ratings per item and 849.0 ratings per user, nearly sixty times higher than the corresponding figure for Amazon Video Games. MovieLens 100k occupies an intermediate position on both metrics.

Table 1: Sparsity statistics of the training sets.

Dataset	Sparsity	Avg. ratings/user	Avg. ratings/item
MovieLens 100k	94.96%	84.8	47.6
Amazon Video Games	99.52%	14.6	4.6
MyAnimeList	87.63%	849.0	100.6

Figure 3.1 illustrates the distributions of ratings per user and per item between the three datasets. All datasets exhibit long-tail distributions, although having different shapes. In MovieLens 100k and Amazon Video Games, user counts are heavily concentrated near the minimum rating threshold, reflecting high data sparsity. In contrast, MyAnimeList shows a distribution of user ratings tending towards log-normal, with the mode falling between 50 and 75, indicating a highly engaged user base that consistently exceeds the minimum threshold. Because the rating filter is applied strictly to the user dimension, item distributions across all datasets extend toward single-rating minimums, which may degrade the precision of latent factor estimates for tail items.

Crucially, this difference in density directly affects the convergence behavior of the Frank-Wolfe variants. When a dataset is highly sparse, the Linear Minimization Oracle (LMO), leveraged by the FW algorithm, struggles to find large dominant patterns because the underlying signal is too weak and fragmented. This forces the algorithm to slowly piece the matrix together by accumulating a large number of individual rank-1 atoms; as a result, sparser datasets naturally require a much higher-rank representation to achieve accurate predictions. Conversely, in denser datasets like MyAnimeList, the overlapping user preferences allow the LMO to capture massive common latent factors, resulting in a highly compact low-rank approximation.

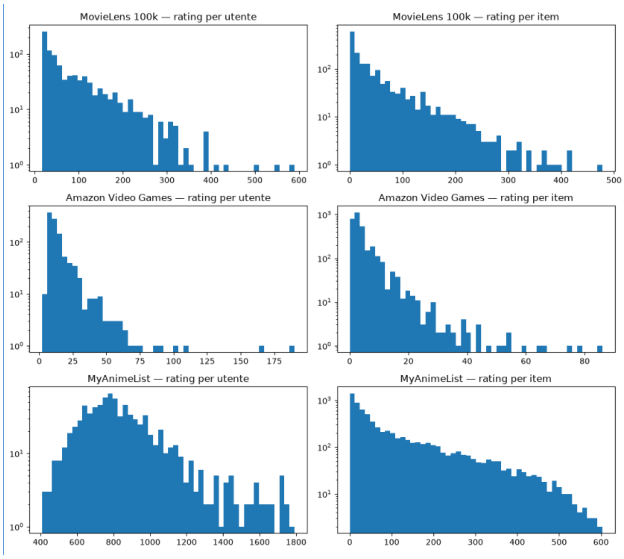


Figure 3.1: Sparsity Analysis: Rating Frequency per User and Item

3.2 Hyperparameter Tuning and Evaluation Metrics

The main hyperparameter that defines the optimization domain is the nuclear norm radius τ. Its value determines the size of the feasible set and therefore controls the trade-off between fitting the observed ratings and favoring a low-rank solution. For each algorithm-dataset pair, a grid search is performed over $\tau \in [4{,}000, 30{,}000]$. The value of τ yielding the lowest RMSE on the validation set is selected for the subsequent evaluation on the test set[5].

Performances are evaluated across three primary dimensions:

- **Convergence Rate:** Monitored via the relative duality gap (g_t/g_0) on both a per-iteration and wall-clock time basis[5].
- **Active Set Dynamics:** Tracking the cardinality of active rank-1 atoms to analyze the memory footprint and the effectiveness of drop steps[5].
- **Out-of-Sample Generalization:** Measured using the Test RMSE evaluated strictly on the unseen hold-out test set[5].

4 The Standard Frank-Wolfe Algorithm

The Standard Frank-Wolfe algorithm is a projection-free iterative method used for solving constrained convex optimization problems[6]. Rather than utilizing expensive projection steps, it linearizes the objective function at the current position and moves toward an extreme point that minimizes this linear approximation over the convex domain[6].

4.1 Sparse Gradients and the Exact Line Search

In the specific context of Matrix Completion, the gradient is non-zero only on the observed entries, and is therefore highly sparse. This sparsity can be exploited to efficiently compute the leading singular vector pair of the gradient using a truncated Singular Value Decomposition. At each iteration, the resulting rank-1 matrix defines an atom, a concept detailed in the next paragraph, that is incorporated into the current approximation, progressively building a low-rank representation of the target matrix.[1]. Furthermore, while the classical FW algorithm typically relies on a pre-defined diminishing step size, such as $\alpha_k = 2/(k+2)$, our implementation utilizes an exact line search[1]. This choice is driven by the quadratic shape of the objective function; in fact, the exact scalar step size $\alpha_k \in [0, 1]$ can be easily computed algebraically, guaranteeing the optimal reduction of the objective function along the chosen descent direction[1].

4.2 Atomic Domains and the Trace Norm LMO

For matrix completion, the optimization domain is the the trace norm ball of radius τ, which is mathematically equivalent to the convex hull of its extreme points[6]. These fundamental rank-1 matrices serve as the extreme points of the domain and are formally defined as *atoms*. Due to this atomic structure, the Linear Minimization Oracle (LMO) simply requires

minimizing the inner product between the gradient and the set of rank-1 atoms[6]. The computational advantage of this projection-free approach is significant. A standard Euclidean projection onto the nuclear norm-ball requires computing a full Singular Value Decomposition (SVD), causing a prohibitive computational complexity of $\mathcal{O}(n_1 n_2 \min\{n_1, n_2\})$ to solve the underlying optimization subproblem[7]. In contrast, as mentioned previously, the Frank-Wolfe LMO completely bypasses this bottleneck by extracting only the top singular vectors via a truncated SVD[5, 7].

4.3 Affine Invariant Curvature

The Frank-Wolfe algorithm is fundamentally affine-invariant: for any reparameterization of the domain D via a linear or affine map, the resulting problem remains identical from the algorithm's perspective. Because of this scale independence, its standard convergence rate relies on the curvature constant C_f, which quantifies the maximum deviation of the objective function from its linear approximation within the feasible domain. This constant is vastly preferred over the standard Lipschitz constant L, as the latter depends on an arbitrarily chosen norm and risks severely overestimating the linearization error. Conversely, C_f seamlessly combines the curvature of the objective function with the specific geometry of the domain D, satisfying the relation $C_f \leq \text{diam}_{\|\cdot\|}(D)^2 L$. In high-dimensional spaces, such as when optimizing quadratic functions over products of probability simplices, L can be exponentially worse than C_f. By bounding this approximation error, the curvature constant mathematically guarantees a sublinear convergence rate of order $\mathcal{O}(1/k)$ for the classical algorithm [5, 6].

To accelerate this standard rate to a linear (geometric) convergence, one must consider the lower bound of the objective function. Standard strong convexity, typically defined by the quadratic term $\frac{\mu}{2}\|x^* - x\|^2$, provides this lower bound but it is not affine-invariant. To by-pass this problem, the geometric strong convexity constant μ_f^A must be introduced. This requires identifying two specific anchor atoms within the generator set $\mathcal{A}$. The standard Frank-Wolfe atom $s_f(x)$, and the away atom $v_f(x)$, which is defined by considering the worst-case scenario among all possible active representations S_x of the current point. We can use them to quantify the alignment between this practical trajectory and the ideal direction toward the optimum x^*:

$$\gamma^A(x, x^*) := \frac{\langle -\nabla f(x), x^* - x \rangle}{\langle -\nabla f(x), s_f(x) - v_f(x) \rangle}$$

By calculating the infimum of this ratio over the entire feasible domain, we derive the fundamental geometric strong convexity constant μ_f^A:

$$\mu_f^A := \inf_{x \in \mathcal{M}} \inf_{\substack{x^* \in \mathcal{M} \\ \langle \nabla f(x), x^* - x \rangle < 0}} \frac{2}{(\gamma^A(x, x^*))^2} (f(x^*) - f(x) - \langle \nabla f(x), x^* - x \rangle)$$

[5]

4.4 The Duality Gap Certificate

To empirically track the $\mathcal{O}(1/k)$ theoretical convergence bound established by the curvature constant, the algorithm relies on a surrogate duality gap, $g(x) := \max_{s \in \mathcal{D}} \langle -\nabla f(x), s - x \rangle$. A critical engineering advantage of Frank-Wolfe is that this gap is computed automatically as a free by-product of solving the LMO[6]. This scalar value acts as a strictly guaranteed certificate for the current approximation quality, continuously upper bounding the unknown primal sub-optimality: $g(x) \geq f(x) - f(x^*)$[6]. By monitoring this gap, the software explicitly verifies that the error shrinks inversely proportional to the iteration count, seamlessly linking the theoretical curvature guarantees to the practical execution of the algorithm[6].

4.5 Limitations: The Zigzagging Phenomenon

While standard Frank-Wolfe ensures a sublinear convergence rate of $\mathcal{O}(1/k)$ for general convex objectives, its performance degrades severely when the optimal solution lies on the boundary of the feasible domain $\mathcal{M}$[5]. Because the Linear Minimization Oracle is mathematically restricted to selecting extreme points (vertices), the algorithm cannot step directly toward a solution located on a flat face[5]. Instead, it is forced to construct the solution by alternating between the vertices defining that face[5]. This results in the "zigzagging" phenomenon, where the search direction becomes nearly orthogonal to the gradient, causing the sublinear convergence rate to stall significantly in practice[5].

5 Advanced Algorithmic Variants

To address the zig-zagging problem of the standard Frank-Wolfe algorithm, several variants introduce mechanisms for modifying the weights assigned to previously selected atoms. In particular, Away-Step Frank-Wolfe (AFW), Pairwise Frank-Wolfe (PFW) algorithm allow the current atomic representation to be actively corrected, and under the assumptions discussed in the paper, these variants achieve a global linear convergence rate over a polytope domain. [5].

5.1 Linear Convergence and Pyramidal Width

Having established the geometric strong convexity constant μ_f^A, and the curvature constant (denoted as C_f^A or C_f) we could use them to mathematically guarantee a linear convergence rate of order $\mathcal{O}((1-\rho)^k)$. Specifically, the sub-optimality $h_t := f(x^{(t)}) - f(x^*)$ decreases according to the relation $h_{t+1} \leq (1 - \rho_f) h_t$ during any "good step", an iteration where the step size γ_t does not reach its maximum allowable limit $\gamma_{\max}$. The exact decay rate ρ_f depends on the algorithmic variant: for AFW it is $\frac{\mu_f^A}{4C_f^A}$, while for PFW it corresponds to the minimum between 1/2 and $\frac{\mu_f^A}{C_f^A}$ [5]. While μ_f^A and C_f^A remain abstract and affine-invariant by construction, Lacoste-Julien and Jaggi [5] demonstrated that a concrete (though no longer affine-invariant) bound on the convergence rate can be obtained using traditional geometric measures. For the strongly convex setting considered, the suboptimality decrease can be

geometrically rewritten as:

$$h_{t+1} \leq (1-\rho)h_t, \qquad \rho = \frac{\mu}{4L}\left(\frac{\delta}{M}\right)^2,$$

where L/μ characterizes the conditioning of the objective function, M denotes the diameter of the feasible domain, and δ is the *pyramidal width* of the atom set, that measures how well the structure of the domain supports feasible descent directions. Consequently, this formulation reveals that the convergence rate depends not only on the properties of the objective function, but fundamentally on the geometry of the feasible domain.

Finally, to establish the global convergence rate across all iterations, one must account for the impact of 'bad steps' where the sub-optimality does not strictly decrease. In the AFW variant, the number of drop steps complete eliminations of a vertex is bounded by half of the total iterations ($t/2$), which guarantees a global decay proportional to $\exp(-\frac{1}{2}\rho_f t)$. Conversely, the PFW variant suffers from a significantly weaker global guarantee: theory demonstrates that up to $3|\mathcal{A}|!$ problematic swap steps can occur between two consecutive good steps. This creates a massive combinatorial bound that makes its theoretical mathematical guarantee quite loose [5].

5.2 Away-Step Frank-Wolfe (AFW)

The standard Frank-Wolfe algorithm can exhibit sublinear convergence when the optimal solution lies on the boundary of the feasible domain, a behavior associated with the previously mentioned zig-zagging phenomenon [5]. The Away-Step Frank-Wolfe (AFW) algorithm addresses this limitation by introducing the possibility of moving away from previously selected atoms. In order to achieve this, AFW maintains an **active set** $\mathcal{S}^{(t)}$, which contains the atoms with strictly positive weights in the current convex decomposition of $x^{(t)}$ [5]. At each iteration, the algorithm uses the LMO to identify the FW atom s_t and, at the same time, selects an away atom

$$v_t \in \arg\max_{v \in \mathcal{S}^{(t)}} \langle \nabla f(x^{(t)}), v \rangle.$$

The corresponding away direction is then defined as

$$d_t^A = x^{(t)} - v_t.$$

AFW compares the potential descent associated with the FW and away directions and selects the one that yields the greatest expected linear decrease. When the away direction is chosen,the weight of the selected away atom is reduced. The maximum feasible step size is

$$\gamma_{\max} = \frac{\alpha_{v_t}}{1-\alpha_{v_t}},$$

which guarantees that the updated point remains a convex combination of the active atoms and therefore remains inside the feasible domain [5]. A **drop step** occurs when the line search selects $\gamma_t = \gamma_{\max}$. In this case, the weight of v_t becomes zero and the atom is removed from the active set. This mechanism allows AFW to eliminate previously selected atoms that are no longer useful, distributing its weight among all the atoms present in the active set. Under the assumptions established by Lacoste-Julien and Jaggi, AFW achieves a global linear convergence rate when optimizing a strongly convex function over a *polytope*[5].

5.3 Pairwise Frank-Wolfe (PFW)

The Pairwise Frank-Wolfe (PFW) variant modifies the standard Frank-Wolfe update rule by directly transferring the weight from an away atom to the new selected FW atom [5]. Therefore, instead of separately choosing between a FW and an away direction as in the AFW, PFW uses the pairwise direction[5]:

$$d_k^{PFW} = s_k - v_k,$$

where s_k is the atom returned by the LMO and v_k is the selected atom from the active set. Therefore, the update changes the weight from v_k to s_k, while keeping the weights of the remaining active atoms unchanged[5]. Since the weight of v_k cannot become negative, the maximum feasible step size is

$$\gamma_{\max} = \alpha_{v_k}.$$

A **drop step** occurs when the line search selects $\gamma_k = \gamma_{\max}$ and the weight of v_k becomes zero, resulting in a reduction of the active set[5]. In contrast, a **swap step** occurs when $\gamma_k = \gamma_{\max}$ but the newly introduced atom s_k replaces v_k, leaving the cardinality of the active set unchanged[5]. These swap steps are one of the main difficulties in the theoretical analysis of PFW and contribute to its weaker worst-case convergence guarantee relative to the other variants[5].

6 Results and Convergence Analysis

The plots regarding the duality gap (Figures 6.1, 6.2, 6.3) confirm the same hierarchy across all three datasets: standard FW converges fastest and to the lowest final gap, followed by Away-Step FW and then Pairwise FW. Comparing the "vs. Iterations" and "vs. Time" plots for MyAnimeList (Figure 6.3) reveals an interesting feature: while Away-Step FW (green) and Pairwise FW (orange) follow fairly similar trajectories in terms of iterations during the first half of the plot, the gap widens enormously in terms of time—Pairwise FW takes nearly 350 seconds, whereas Away-Step FW takes around 120 seconds for a comparable number of iterations. This confirms that the swap steps in Pairwise FW entail a much higher computational cost per iteration [5]. Regarding MovieLens 100k (Figure 6.2): the three curves are very similar for most of the execution, with Pairwise FW diverging (performing worse) only in the final iterations. This suggests that the relative advantage or disadvantage of the variants depends heavily on the specific dataset structure—differences between algorithms are marginal for MovieLens, whereas for Amazon Video Games (Figure 6.1), they are distinct from the very first iterations.

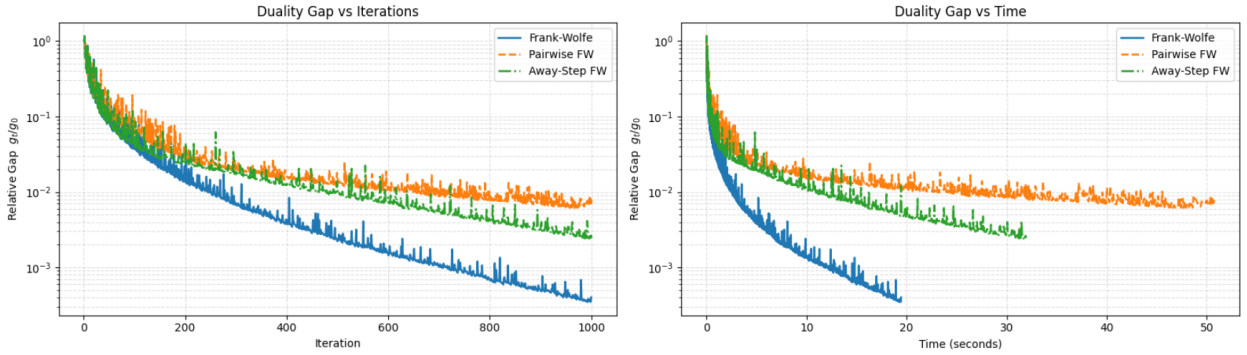


Figure 6.1: Convergence Analysis - Amazon Video Games

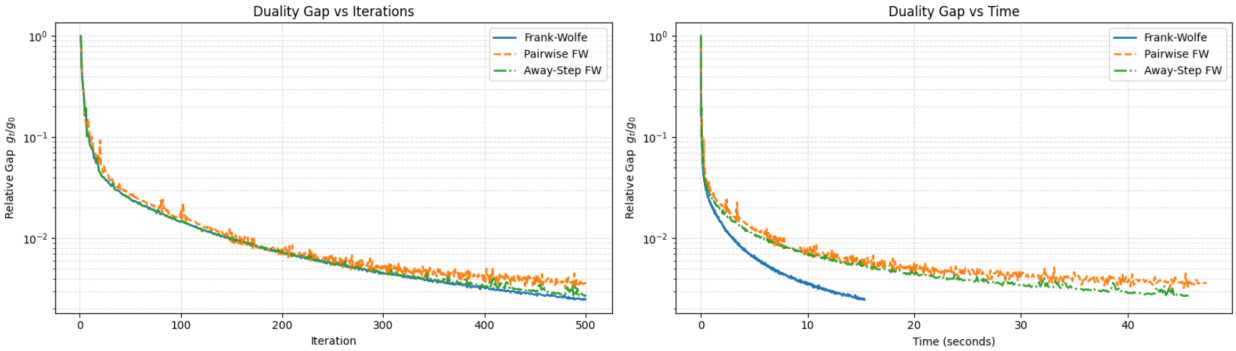


Figure 6.2: Convergence Analysis - Movielens 100k

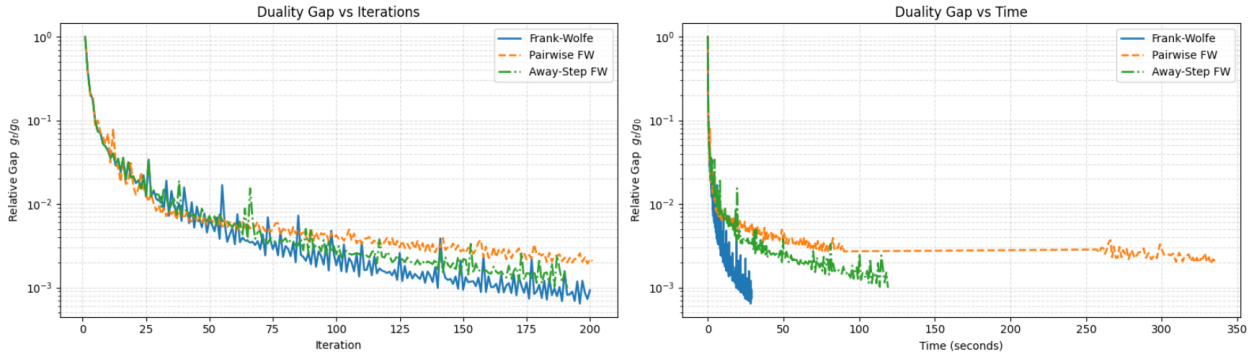


Figure 6.3: Convergence Analysis - MyAnimeList

6.1 Active Set Dynamics

Figure 6.4 illustrates the phenomenon of active-set bloat, caused by the fact that the domain is not polyhedral.

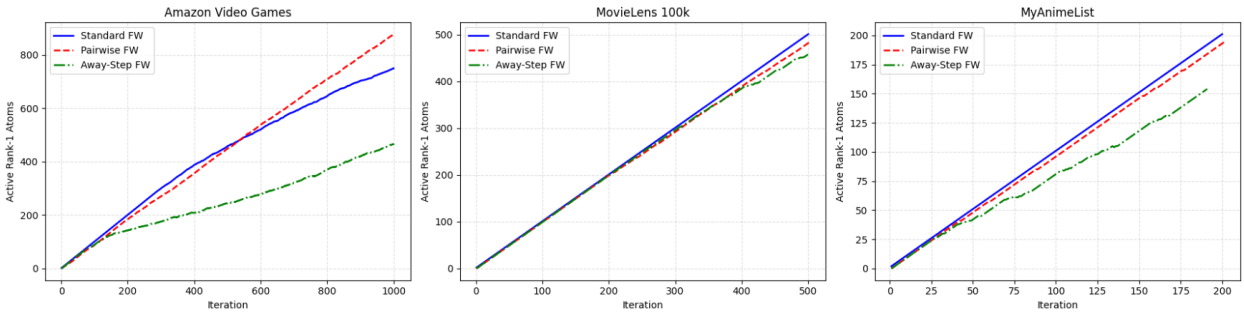


Figure 6.4: Active set bloat on nuclear norm

It follows that the linear convergence guarantee for AFW/PFW no longer holds [5], as it applies only to polyhedral domains with a finite number of atoms/vertices; consequently, the "drop step" mechanism intended to keep the active set under control lacks the guaranteed effectiveness it would have on a true polytope.
Looking at the specifics, for the MovieLens 100k dataset, the three curves almost overlap, showing nearly identical linear growth across all three variants. Here, "bloat" is minimal—even Away-Step FW fails to distinguish itself significantly from standard FW in terms of active set control.
A more pronounced difference emerges among the three variants with the Amazon Video Games dataset. Standard FW and Pairwise FW grow almost linearly up to 750–880 atoms, whereas Away-Step FW grows noticeably more slowly, plateauing at around 470. The drop-step mechanism works partially here, reducing bloat by approximately 40% compared to standard FW. In the case of MyAnimeList, however, the Away-Step FW curve (green) exhibits a "stepped" pattern followed by steeper ascents. Indeed, the algorithm alternates between phases where it actively removes atoms and phases where it must add new ones (ascents).

We can therefore conclude neither AFW nor PFW succeeds in achieving effective active-set control comparable to the polytope case on a domain such as the nuclear norm. This empirical result confirms that the linear convergence framework of Lacoste-Julien & Jaggi presupposes a polytope; the "active set bloat" observed here is the practical manifestation of this lack of a formal guarantee.

6.2 Convergence Efficiency

The efficiency gap observed in the plots stems from two independent factors: the number of iterations required to reach a given gap ("vs Iterations") and the cost in terms of time per iteration (the ratio between the two panels: "vs Time" divided by "vs Iterations").

In the timing plots (Figure 6.1, 6.2, 6.3), AFW is consistently slower than standard FW for a given iteration; the cost of calculating the "away" direction results in a higher time per iteration.
On MyAnimeList, for instance, it reduces active set bloat (Figure 6.4), but at a time cost that remains significantly higher than that of standard FW (Figure 6.3 120s versus 20–30s

for FW). Searching for the worst atom requires computing the inner product between the sparse gradient matrix and every single atom currently stored in the active set $\mathcal{S}^{(t)}$. When the active set bloats to hundreds of atoms on the Nuclear Norm ball, this linear search through the active set becomes a massive computational bottleneck, completely destroying the time efficiency per iteration.
In any case, Pairwise FW remains the least efficient: the active set grows almost as it does with standard FW, and the cost per iteration is the highest. This is particularly evident on MyAnimeList, where the total time (∼340s) is more than ten times that of standard FW, resulting in a final gap that remains the largest.

6.3 Out-of-Sample Performance

The bar chart (Figure 6.5) and the table (Figure 6.6) make the comparison clearer. Standard FW consistently generalizes better across all three datasets.

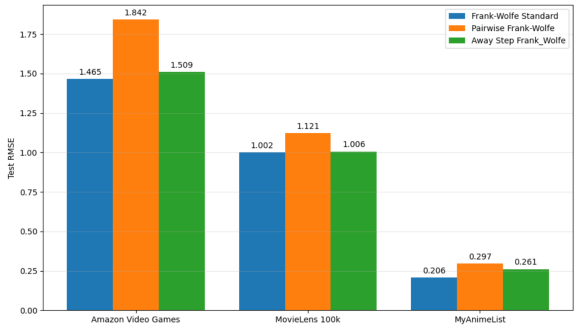


Figure 6.5: Out-of-Sample Performance

Dataset	FW Standard	Pairwise FW	Away-Step FW	Δ (PFW vs FW)
Amazon Video Games	1.465	1.842	1.509	+25.7%
MovieLens 100k	1.002	1.121	1.006	+11.9%
MyAnimeList	0.206	0.297	0.261	+44.2%

Figure 6.6: Summary Table

The largest relative gap is observed on MyAnimeList (+44.2% RMSE for PFW compared to FW), a somewhat counterintuitive result given that PFW's active set grows almost as much as that of standard FW (Figure 6.4), showing none of the containment that AFW achieves through its drop-step mechanism on this dataset. A plausible explanation is that, despite this similarity in active-set size, the repeated swap steps continuously reallocate weight between the FW and away atoms without ever pruning the active set; this ongoing reallocation can produce less stable intermediate solutions during training, resulting in weaker generalization. Standard FW, by contrast, adds atoms in a monotonic and "cautious" manner, producing a smoother learning trajectory and, consequently, better generalization performance in this case.

7 Conclusion

This report investigated the application of the Frank-Wolfe algorithm and its advanced variants to the Matrix Completion problem in recommender systems. While Away-Step and Pairwise Frank-Wolfe offer theoretical global linear convergence, our empirical results

demonstrate that these guarantees rely on the assumption of a polyhedral optimization domain.

When applied to the continuous, curved geometry of the nuclear norm ball, the advanced variants suffer from severe active set bloat and significant computational cost. Because the domain lacks finite, flat faces, the algorithms struggle to trigger effective drop steps, nullifying their primary structural advantage. Consequently, the Away-Step and Pairwise variants not only fail to converge faster in terms of time, but each also appears to harm out-of-sample generalization through a distinct mechanism: for Away-Step FW, the repeated shrinking and re-expansion of the active set produces less stable intermediate solutions; for Pairwise FW, the inability to effectively prune the active set—combined with continuous mass reallocation via swap steps—likely has a similar destabilizing effect.

Ultimately, the Standard Frank-Wolfe algorithm proved to be the most robust and efficient approach for this specific application. These findings highlight a crucial lesson in optimization for data science: theoretically sophisticated variants do not automatically translate to practical superiority when the underlying geometric assumptions of the domain are violated.

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